

# Long Term Expectations and Aggregate Fluctuations

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# Overview

Observed volatility in both stock markets and aggregate economic activities.

- Shiller (1981)'s volatility puzzle
- What's driving excessive volatility in the stock market? Does the same driver also affect real activity?
- This paper: all linked by long-term expectations
- Main variable: analysts' consensus forecast of a firm's long-term earnings growth (LTG), aggregated across firms in S&P500
- Aligns with Keynes' "animal spirits" intuition: changing business conditions → excessive changes in expectations → amplify shocks

[Related Literature I](#)[Lit II](#)[Lit III](#)

# Overview

## 1. Shiller's Volatility Puzzle

**This paper: measure long-term expectations using LTG  
to explain volatility of **financial** and **macro** variables:**

2. Expectations-based Stock Price
3. LTG and Interest Rates / Credit  
Spreads
4. LTG and Investment
5. LTG and the Business Cycle

# Overview

## 1. Shiller's Volatility Puzzle

**This paper: measure long-term expectations using LTG  
to explain volatility of **financial** and **macro** variables:**

### 2. Expectations-based Stock Price

### 3. LTG and Interest Rates / Credit Spreads

### 4. LTG and Investment

### 5. LTG and the Business Cycle

(i) LTG accounts for the excess volatility  
puzzle and other asset prices

(ii) LTG sheds light on the dynamics of  
investment and other business cycle  
indicators, including investment shocks

(iii) Non-rationality, as measured by predictable forecast errors, plays a key role

# Roadmap

1. Motivation: Shiller's Volatility Puzzle
2. Expectations-Based Price Index
3. LTG and the Financial Cycle
4. LTG and Boom-Bust Investment Cycles
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# Shiller's Volatility Puzzle

Campbell and Shiller (1987, 1988) express the price-dividend ratio of a stock by

$$\begin{array}{c}
 p_t - d_t \\
 \downarrow \\
 \text{log price-dividend ratio}
 \end{array}
 = \frac{k}{1 - \alpha} + \underbrace{\sum_{s \geq 0} \alpha^s g_{t+1+s}}_{\text{dividend growth}} - \underbrace{\sum_{s \geq 0} \alpha^s r_{t+1+s}}_{\text{realized return}} \quad (1)$$

where

- $p_t$  is the log price and  $d_t$  is the log dividend at time  $t$
- $g_{t+1+s} = d_{t+1+s} - d_{t+s}$ : dividend growth from  $t + s$  to  $t + s + 1$
- $r_{t+1+s}$ : realized stock return
- $k$  is a constant, and  $\alpha = e^{pd} / (1 + e^{pd})$  depends on the average log price dividend ratio  $pd$ .

# Shiller's Volatility Puzzle

- Under rationality and a constant required return  $r$ , the stock price is given by

$$p_t^R = d_t + \frac{k - r}{1 - \alpha} + \sum_{s \geq 0} \alpha^s E_t(g_{t+1+s}) \quad (2)$$

## Intuition

- Weighted average of dividend growth on the RHS should be less volatile than realized dividend growth, which has low volatility itself
- However, even volatility in realized  $g_{t+1+s}$  cannot account for the large observed volatility of  $p_t$

# Rational Price: Assumptions and Results

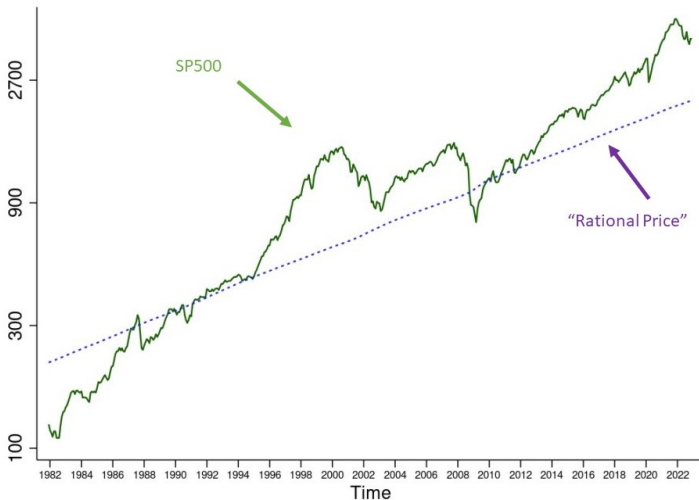
- Shiller: assumed perfect foresight of future dividends and a value of the rational stock price in the last period.

$$p_t^* = d_t + \frac{1 - \alpha^{T-t}}{1 - \alpha}(k - r) + \sum_{s=t}^T \alpha^{s-t}(d_{s+1} - d_s) + \alpha^{T-t}(p_{2022}^* - d_{2022}) \quad (3)$$

- Here: replicate Shiller's exercise using earnings over 1981-2022
  - ▶ Terminal EPS  $D_{2022} = 66.92$ , terminal log price  $p_{2022}^* = \ln\left(\frac{D_{2022}}{r-g}\right)$
  - ▶  $r = 8.75\%$ ,  $g = 5.79\%$  (sample average)
  - ▶  $\alpha = 0.9981$  (monthly frequency)
  - ▶  $k = -\log(\alpha) - (1 - \alpha)\log\left(\frac{1}{\alpha-1}\right) = 0.0138$ .
  - ▶ Little change if dividends were used as  $d_t$  instead of EPS



# SP500 vs Rational Price (Log Scale): 1981-2022



# Common Approaches to Account for Stock Price Volatility

$$p_t^R = d_t + \frac{k - r}{1 - \alpha} + \sum_{s \geq 0} \alpha^s E_t(g_{t+1+s})$$

- Most asset pricing research focus on the discount rate  $r$ : construct theories of investor preferences that admit variation in the price and quantity of risk
- Behavioral Finance: mostly focused on extrapolative expected returns (e.g. Barberis et al. 2015, 2018, Hirshleifer et al. 2015, etc.)
- A smaller body of work relaxed the assumption of rational expectations of dividends, most recently BGLS (2019, 2022): replace  $E_t[g_{t+1+s}]$  with  $\tilde{E}_t[g_{t+1+s}]$ 
  - ▶ Falls under the broad agenda on diagnostic expectations

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# Measured Expectations of Future Fundamentals

Data: monthly analyst forecasts for S&P500 firms from IBES Unadjusted US Summary Statistics

- Use earnings forecasts instead of dividends to expand data coverage
- Median forecasts of a firm's earnings per share ( $EPS_{it}$ , starting from 3/1976) and long-term earnings growth ( $LTG_{it}$ , starting from 12/1981)
- Firm level estimates are then aggregated into one and two year ahead  $EPS_t$  forecasts and  $LTG_t$  (up to  $t = 10$ ). [LTG Aggregation](#) [Data Description](#)
- LTG: expected annual increase in operating earnings over the company's next full business cycle (referring to a period of 3-5 years)

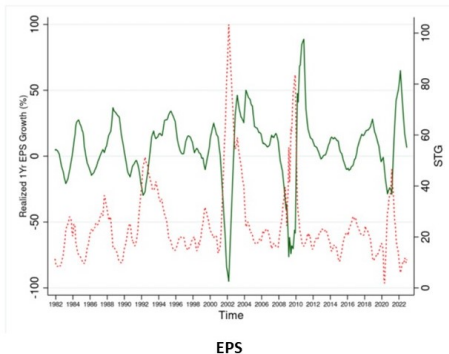
# Volatility of Earnings Growth and Expectations

- Short-term expectations move mainly due to short-term mean reversion of earnings growth
- LTG captures persistent fluctuations in estimated growth potential

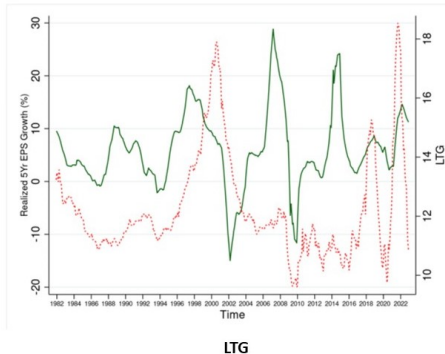
[Data Summary](#)

STG

LTG



EPS



LTG

Red: STG / LTG; Green: realized 1-year / 5-year earnings growth

# LTG: Potential Concerns

## 1. Analysts may distort their forecasts due to agency

- ▶ Arguably stable and hence unlikely to affect time series variation, especially for *S&P* firms that are followed by all brokerage houses
- ▶ Using median forecasts reduce impact of outliers
- ▶ BGLS (2019): LTG respond to news (high LTG forecasts after a sequence of positive surprises)

## 2. Analysts estimate expected earnings growth using stock prices themselves

- ▶ BGLS (2022) found strong evidence against this: LTG revisions are more reliably explained by past earnings growth than by past returns, at both aggregate and firm level
- ▶ LTG predicts stock returns even after controlling for P-E ratio

Conclusion: LTG as a valuable proxy for market beliefs about future fundamentals

# The Expectations Based Index

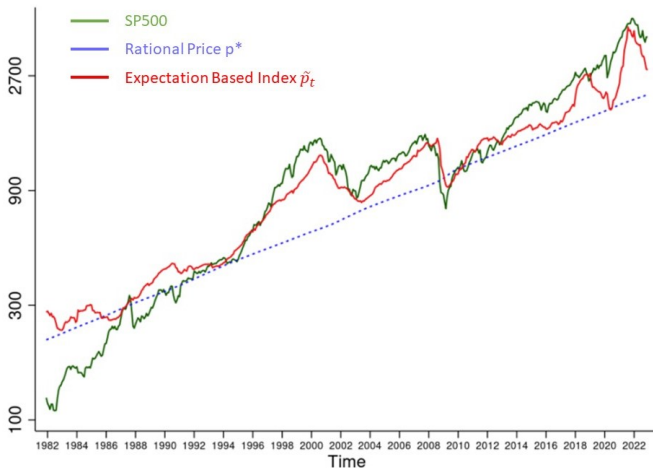
$$\tilde{p}_t = e_t + \frac{\tilde{k} - r}{1 - \alpha} + \ln \left( \frac{\text{EPS}_{t,t+1}}{\text{EPS}_t} \right) + \alpha \ln \left( \frac{\text{EPS}_{t,t+2}}{\text{EPS}_{t,t+1}} \right) + \sum_{s=2}^{10} \alpha^s \text{LTG}_t + \frac{\alpha^{10}}{1 - \alpha} g \quad (4)$$

Key difference: use of expectations data

- Use expected EPS growth between  $t$  and  $t + 2$
- Use  $\text{LTG}_t$  to capture expected earnings growth between  $t + 3$  to  $t + 10$  (average duration of a business cycle in their data is 10 years)
- Set expected growth rate after  $t + 10$  to be  $g = 3.73\%$  (to match the average growth rates of  $\tilde{p}_t$  and  $p_t$ )

# The Expectations Based Index $\tilde{p}_t$

- Captures low frequency price movements well



Adjusted for Inflation



# Volatility of Log Price Changes

- Delivers a very realistic amount of price volatility (SD of one-year changes in log price), much higher than the rational benchmark

**Table 1:** Volatility of Log Price Changes

|                       | Earnings Index |              |                    |
|-----------------------|----------------|--------------|--------------------|
|                       | $\Delta p$     | $\Delta p^*$ | $\Delta \tilde{p}$ |
| <i>Variance</i>       | 15.7%          | 0.7%         | 15.3%              |
| <i>Conf. interval</i> | 14.7-16.7%     | 0.6-0.7%     | 14.4-16.3%         |

Note: The table reports the standard deviation and 95<sup>th</sup> confidence interval of one-year change in: (a) the log of the price of the S&P500 index,  $\Delta p$ , (b) the rational benchmark index,  $\Delta p^*$  (equation 3), and (c) the price index based on earnings forecasts (equation 4),  $\Delta \tilde{p}$ . The sample period is 12/1982 to 12/2022.

# Rationality Test 1: Using LTG to Predict Forecast Errors

- The LHS  $y_{t+h}$  is the prediction error at time  $t + h$  of 1,2,5-year ahead earnings
- High LTG captures periods of excess aggregate optimism: LTG systematically predicts positive forecast errors and thus future disappointment of earnings growth.
- In contrast, short-term expected growth do not predict forecast errors. STG

Time Horizon (h) of Dependent Variable (Quarters)

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|----|
|---|---|---|---|---|---|---|---|---|---|----|

Panel B: Estimates From  $y_{t+h} = B_h LTG_t + \varepsilon_{t+h}$   
Independent Variable:  $LTG_t$

Dependent Variable

|                                                  |                    |                     |                     |                     |                     |                     |                     |                    |                    |                    |                   |
|--------------------------------------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|--------------------|--------------------|--------------------|-------------------|
| $y_{t+h} = STG1_{t+h} - \Delta_4 e_{t+h+4}$      | 9.99***<br>[ 2.88] | 12.58***<br>[ 2.53] | 13.82***<br>[ 2.14] | 13.80***<br>[ 2.09] | 13.21***<br>[ 2.06] | 12.25***<br>[ 2.03] | 11.15***<br>[ 2.01] | 9.67***<br>[ 2.11] | 7.47***<br>[ 2.23] | 5.26**<br>[ 2.36]  | 3.35<br>[ 2.39]   |
| $y_{t+h} = STG2_{t+h} - \Delta_8 e_{t+h+8}/2$    | 5.36***<br>[ 1.40] | 5.58***<br>[ 1.50]  | 5.53***<br>[ 1.71]  | 5.23***<br>[ 1.95]  | 4.18**<br>[ 1.97]   | 3.42<br>[ 2.15]     | 1.96<br>[ 1.93]     | 0.66<br>[ 1.67]    | -0.36<br>[ 1.68]   | -1.18<br>[ 1.69]   | -2.12<br>[ 1.46]  |
| $y_{t+h} = LTG_{t+h} - \Delta_{20} e_{t+h+20}/5$ | 3.69***<br>[ 0.74] | 3.49***<br>[ 0.74]  | 3.04***<br>[ 0.75]  | 2.38***<br>[ 0.78]  | 1.53*<br>[ 0.82]    | 0.58<br>[ 0.86]     | -0.33<br>[ 0.90]    | -1.14<br>[ 0.90]   | -1.63*<br>[ 0.87]  | -1.81**<br>[ 0.85] | -1.69*<br>[ 0.87] |

Forecast errors  $y_{t+h}$  are in percentage terms. These findings align with BGLS (2022).

BGLS 2022

## Rationality Test 2: LTG and Expected Stock Returns

- Rational models: in good times, investors require and expect lower returns
- Here: periods of high LTG exhibit high optimism and not low required returns

**Table 2.** LTG, Forecast Errors and Expectations of Stock Returns

| Dependent Variable                                       | Time Horizon (h) of Dependent Variable (Quarters)                                                                 |                  |                |                |                |                |                 |                   |                   |                 |                |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------|----------------|----------------|----------------|----------------|-----------------|-------------------|-------------------|-----------------|----------------|
|                                                          | 0                                                                                                                 | 1                | 2              | 3              | 4              | 5              | 6               | 7                 | 8                 | 9               | 10             |
|                                                          | Panel B: Estimates From $y_{t+h} = B_h LTG_t + \mathbf{X}_t + \varepsilon_{t+h}$<br>Independent Variable: $LTG_t$ |                  |                |                |                |                |                 |                   |                   |                 |                |
| $y_{t+h} = \text{Expected 1Y sp500 return } (cfo)_{t+h}$ | 0.36<br>[0.25]                                                                                                    | 0.61**<br>[0.25] | 0.45<br>[0.31] | 0.43<br>[0.34] | 0.34<br>[0.37] | 0.25<br>[0.43] | -0.38<br>[0.25] | -0.75**<br>[0.28] | -0.61**<br>[0.27] | -0.19<br>[0.30] | 0.09<br>[0.27] |

- Controls: 12 lags of dependent variable
- Similar effects of LTG on other measures of expected returns
- Granger test supports the view that LTG drives expected returns, not the reverse.

Granger Test

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# An Asset Pricing Model

Modify a standard model to allow for non-rational, overreacting beliefs about fundamentals to link LTG to interest rates and spreads:

- Endowment economy with output growth:

$$g_{t+1} = \mu g_t + v_{t+1} \quad (5)$$

- Instead, investors use an incorrect model of output growth:

$$\tilde{g}_{t+1} = \mu g_t + \omega_t + v_{t+1} \quad (6)$$

- Time-varying belief distortions  $\omega_t$ :

- ▶  $\omega_t > 0$ : excessive optimism
- ▶ Persistent and compounds reactions to present and past news

$$\omega_t = \rho \omega_{t-1} + \theta v_t \quad (7)$$

where  $\theta > 0$  implies beliefs over-react

- ▶ BGLS (2022): special case of the diagnostic expectations model when  $\theta > 0$

# Asset Pricing Model: Representative Agent

- Representative consumer with CRRA utility and risk aversion parameter  $\gamma$
- Asset prices are set according to FOC:

$$\tilde{E}_t[R_{t+1}\beta(1 + g_{t+1})^{-\gamma}] = 1 \quad (8)$$

- This can be rewritten as

$$E_t[R_{t+1}\beta(1 + g_{t+1})^{-\gamma}M(g_{t+1}, g_t, \omega_t)] = 1 \quad (9)$$

where  $M(g_{t+1}, g_t, \omega_t)$  captures the investor's belief distortions

- Replaces non-standard preferences but is not observationally equivalent to them: shifts in beliefs can be disciplined using the expectations data
- This is just a minimalistic model of belief overreaction

Quantitative Drawbacks

## Equilibrium Risk-free Rate

- Assuming joint lognormality of returns and fundamentals, the Euler equation pins down the equilibrium risk free rate and risk premium.

$$r_{t+1}^f = -\log \beta - \frac{1}{2}\gamma^2\sigma_g^2 + \gamma(\mu g_t + \omega_t) \quad (10)$$

- New term  $\omega_t$ : during times of excessive optimism about future growth, consumer is reluctant to save / more willing to borrow, so  $r^f$  is higher.
- Two predictions:** High optimism (proxied by upward revisions of  $LTG_t$ ) should be associated with
  - higher current interest rate  $r_{t+1}^f$
  - reversal of interest rates  $r_{t+s}^f$  in the future: mean reversion in output growth if  $\mu < 1$  or disappointment of future  $\omega_t$  since  $\rho < 1$ .

# Equilibrium Risk Premium

- Similarly, assuming joint lognormality of returns and fundamentals:

$$E_t(r_{t+1}) - r_{t+1}^f = \left(\gamma - \frac{\omega_t}{\sigma_g^2}\right) \sigma_{rg} \quad (11)$$

- New term  $\omega_t$ : when the consumer becomes more optimistic about future growth, the risk premium is persistently low.
- **Two predictions:** higher current optimism (upward revisions of *LTG*) should
  - ▶ be associated with low contemporaneous credit spreads
    - ★ Good current news leads to upward revision: high bond price and high realized excess returns
  - ▶ predict higher credit spreads in the future
    - ★ Also low bond price and low average realized excess return in the future



# Estimate of LTG Revision on Asset Prices

- Quarterly local projections: what is the impact of a 1 SD change in  $\Delta_4 LTG_t$  (yearly  $LTG_t$  change) on year-on-year percentage changes in  $y_t$ ?
  - ▶  $y_t$ : 1-year T-bill rate, 10-year T-bill rate, Baa credit spread

$$\Delta_4 y_{t+h} = B_h \Delta_4 LTG_t + \mathbf{X}_t + \varphi_{t+h}, \quad h = 0, 1, \dots, 10$$

- Controls  $\mathbf{X}_t$ : 12 lags of: the dependent variable, yearly changes in the policy rate, yearly CPI inflation, yearly log change in SP500
  - ▶ Assuage concerns that LTG shock may capture fundamental mean reversion, monetary policy response, and potentially time varying required return embodied in stock valuations

[Data Description](#)[Data Summary](#)

# Estimate of LTG Revision on Asset Prices

- Consistent with Equation (10): an increase in optimism is associated with higher interest rates at  $h = 0, 1, 2, 3$ , followed by a period of stability. After 6 quarters, rates revert and decline

|                                                                | Time Horizon (h) of Dependent variable |                    |                    |                    |                 |                  |                     |                     |                     |                     |                     |
|----------------------------------------------------------------|----------------------------------------|--------------------|--------------------|--------------------|-----------------|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                                                                | 0                                      | 1                  | 2                  | 3                  | 4               | 5                | 6                   | 7                   | 8                   | 9                   | 10                  |
| Panel A: dependent variable $\Delta_4 \text{tbill } 1y_{t+h}$  |                                        |                    |                    |                    |                 |                  |                     |                     |                     |                     |                     |
| $\Delta_4 LTG_t$                                               | 0.21***<br>[ 0.07]                     | 0.40***<br>[ 0.07] | 0.44***<br>[ 0.09] | 0.39***<br>[ 0.12] | 0.12<br>[ 0.13] | -0.19<br>[ 0.13] | -0.37***<br>[ 0.13] | -0.49***<br>[ 0.12] | -0.62***<br>[ 0.13] | -0.74***<br>[ 0.15] | -0.82***<br>[ 0.17] |
| N                                                              | 151                                    | 151                | 151                | 151                | 151             | 151              | 151                 | 151                 | 151                 | 151                 | 151                 |
| AR <sup>2</sup>                                                | 0.85                                   | 0.66               | 0.48               | 0.25               | 0.17            | 0.24             | 0.33                | 0.38                | 0.35                | 0.30                | 0.24                |
| Panel B: dependent variable $\Delta_4 \text{tbill } 10y_{t+h}$ |                                        |                    |                    |                    |                 |                  |                     |                     |                     |                     |                     |
| $\Delta_4 LTG_t$                                               | 0.18**<br>[ 0.07]                      | 0.35***<br>[ 0.08] | 0.41***<br>[ 0.08] | 0.40***<br>[ 0.09] | 0.16<br>[ 0.12] | -0.09<br>[ 0.12] | -0.24**<br>[ 0.10]  | -0.32***<br>[ 0.11] | -0.32***<br>[ 0.12] | -0.40***<br>[ 0.12] | -0.48***<br>[ 0.13] |
| N                                                              | 151                                    | 151                | 151                | 151                | 151             | 151              | 151                 | 151                 | 151                 | 151                 | 151                 |
| AR <sup>2</sup>                                                | 0.77                                   | 0.60               | 0.49               | 0.37               | 0.25            | 0.27             | 0.30                | 0.29                | 0.24                | 0.20                | 0.16                |

# Estimate of LTG Revision on Asset Prices

- Growing optimism about future earnings growth (e.g. due to high current growth) is associated with lower contemporaneous spreads and a predictable tightening of credit market starting from quarter 5

|                                                                               | Time Horizon (h) of Dependent variable |                    |                   |                  |                 |                  |                   |                   |                   |                  |                 |
|-------------------------------------------------------------------------------|----------------------------------------|--------------------|-------------------|------------------|-----------------|------------------|-------------------|-------------------|-------------------|------------------|-----------------|
|                                                                               | 0                                      | 1                  | 2                 | 3                | 4               | 5                | 6                 | 7                 | 8                 | 9                | 10              |
| Panel C: dependent variable<br>$\Delta_4 \text{baa credit spread } 10y_{t+h}$ |                                        |                    |                   |                  |                 |                  |                   |                   |                   |                  |                 |
| $\Delta_4 LTG_t$                                                              | -0.10<br>[ 0.07]                       | -0.13**<br>[ 0.06] | -0.12*<br>[ 0.06] | -0.08<br>[ 0.07] | 0.08<br>[ 0.09] | 0.19*<br>[ 0.11] | 0.23**<br>[ 0.10] | 0.22**<br>[ 0.09] | 0.19**<br>[ 0.09] | 0.16*<br>[ 0.09] | 0.12<br>[ 0.10] |
| N                                                                             | 151                                    | 151                | 151               | 151              | 151             | 151              | 151               | 151               | 151               | 151              | 151             |
| AR <sup>2</sup>                                                               | 0.74                                   | 0.55               | 0.42              | 0.28             | 0.19            | 0.22             | 0.23              | 0.18              | 0.07              | -0.03            | -0.06           |

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# LTG and Boom-Bust Investment Cycles

- Keynes: Expectations of long term profits ("animal spirits") is an important source of volatility in financial markets and are connected to firms' desire to invest
- Here: reconcile financial and business cycle volatility by studying the connection between LTG and real investment

# Estimate of Change in LTG on Changes in Investment

- Estimate local projections for aggregate year-on-year change in investment with the same controls as before

$$\Delta_4 \text{investment-to-capital}_{t+h} = B_h \Delta_4 \text{LTG}_t + \mathbf{X}_t + \epsilon_t$$

where  $\Delta_4 \text{investment-to-capital}_t$  is the 4-quarter log change in the ratio of non-residential investment to the previous year's cost of capital stock

[Data Summary](#)

- Sizable increase peaking at 3% in the year after (0.4 SD of YoY investment growth)

|                                                                                                                              | Time Horizon of Dependent Variable<br>(Quarters) |                    |                    |                    |                    |                 |                  |                     |                     |                     |                  |
|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------|--------------------|--------------------|--------------------|-----------------|------------------|---------------------|---------------------|---------------------|------------------|
|                                                                                                                              | 0                                                | 1                  | 2                  | 3                  | 4                  | 5               | 6                | 7                   | 8                   | 9                   | 10               |
| Panel A: Estimates From<br>$\Delta_4 \text{investment-to-capital}_t = B_h \Delta_4 \text{LTG}_t + \mathbf{X}_t + \epsilon_t$ |                                                  |                    |                    |                    |                    |                 |                  |                     |                     |                     |                  |
| $\Delta_4 \text{LTG}_t$                                                                                                      | 0.70***<br>[ 0.20]                               | 1.83***<br>[ 0.42] | 2.65***<br>[ 0.50] | 3.21***<br>[ 0.53] | 2.45***<br>[ 0.60] | 0.57<br>[ 0.79] | -1.27<br>[ 0.81] | -2.58***<br>[ 0.74] | -2.63***<br>[ 0.64] | -1.83***<br>[ 0.63] | -0.68<br>[ 0.60] |
| AR <sup>2</sup>                                                                                                              | 0.94                                             | 0.85               | 0.75               | 0.59               | 0.36               | 0.13            | 0.11             | 0.17                | 0.22                | 0.19                | 0.15             |
| N                                                                                                                            | 150                                              | 150                | 150                | 150                | 150                | 150             | 150              | 150                 | 150                 | 150                 | 150              |

# Including Predictable Forecast Errors

Is the long run investment decline connected to the disappointment of optimistic expectations? Does high excess optimism predict future investment busts?

- First stage:

$$FE_t = LTG_t - \Delta_{20}e_{t+20}/5 = \Phi LTG_t + \epsilon_t \rightarrow \hat{FE}_t$$

- ▶  $FE_t$ : forecast error; difference between expected and realized average 5-year aggregate EPS growth from Section 2
- ▶  $\hat{FE}_t$  are fitted values from this regression: predictable component of LTG forecast errors

- Second stage:

$$\Delta_4 \text{investment-to-capital}_{t+h} = B_h \Delta_4 LTG_t + \delta_h \hat{FE}_t + \mathbf{X}_t + \epsilon_t$$

# $\Delta_4 LTG_t$ and $\hat{FE}_t$ on $\Delta_4 \text{Investment-to-Capital}_{t+h}$

- Excess optimism, captured by predictable disappointment, accounts for the entire future reversal in aggregate investment growth: 1 SD increase in  $\hat{FE}_t$  leads to a 2% fall (0.27 SD) in investment growth 2 years later
- Current LTG shock stimulates investment in the near term, just like a good fundamental shock

|                  | Time Horizon of Dependent Variable<br>(Quarters) |                    |                    |                    |                    |                    |                     |                     |                     |                     |                     |
|------------------|--------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                  | 0                                                | 1                  | 2                  | 3                  | 4                  | 5                  | 6                   | 7                   | 8                   | 9                   | 10                  |
| $\Delta_4 LTG_t$ | 0.85***<br>[ 0.31]                               | 1.67***<br>[ 0.49] | 2.20***<br>[ 0.64] | 2.80***<br>[ 0.86] | 2.47***<br>[ 0.89] | 1.47*<br>[ 0.88]   | 0.55<br>[ 0.84]     | -0.24<br>[ 0.76]    | -0.84<br>[ 0.69]    | -0.75<br>[ 0.73]    | -0.24<br>[ 0.82]    |
| $\hat{FE}_t$     | 0.13<br>[ 0.14]                                  | 0.30<br>[ 0.24]    | 0.29<br>[ 0.33]    | 0.07<br>[ 0.43]    | -0.44<br>[ 0.46]   | -1.15**<br>[ 0.47] | -1.70***<br>[ 0.44] | -2.02***<br>[ 0.39] | -1.98***<br>[ 0.36] | -1.80***<br>[ 0.37] | -1.61***<br>[ 0.42] |
| AR <sup>2</sup>  | 0.95                                             | 0.87               | 0.75               | 0.57               | 0.37               | 0.17               | 0.15                | 0.20                | 0.25                | 0.25                | 0.20                |
| N                | 138                                              | 138                | 138                | 138                | 138                | 138                | 138                 | 138                 | 138                 | 138                 | 138                 |

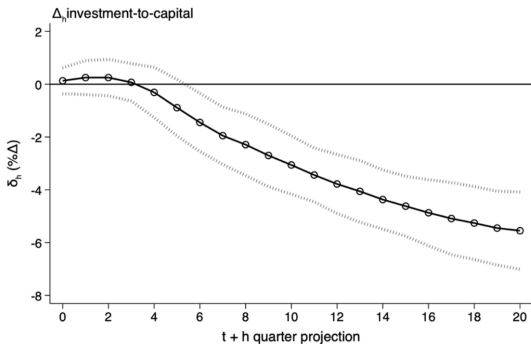


# Impulse Response of Cumulative Investment Growth to $\hat{F}\hat{E}_t$

Impact of 1 SD change in predictable forecast error  $\hat{F}\hat{E}_t$  on  $\Delta_h$  investment-to-capital $_{t+h}$

- Consistent with excessive optimism at  $t$  causing excessive investment in year 1, leading to:
  - (i) Disappointment in expectations going forward
  - (ii) A cutback of inefficient investment in subsequent years

IRF Table



# LTG and Investment with Firm and Time FE

- Column 1: high firm level LTG predicts future disappointment in earnings growth
- Column 2-6: an upward LTG revision is associated with high year-on-year investment in the near term, but a large and predictable decline going forward

Details

|                     | (1)                   | (2)                                                                                           | (3)                    | (4)                    | (5)                    | (6)                    |
|---------------------|-----------------------|-----------------------------------------------------------------------------------------------|------------------------|------------------------|------------------------|------------------------|
|                     | $FE_{it}$             | Estimates From:                                                                               |                        |                        |                        |                        |
|                     |                       | $\Delta_4 i_{i,t+h} = B_h \Delta_4 LTG_{it} + \delta_h \widehat{FE}_{it} + \varepsilon_{t+h}$ |                        |                        |                        |                        |
|                     |                       | $h = 0$                                                                                       | $h = 6$                | $h = 12$               | $h = 18$               | $h = 24$               |
| $LTG_{it}$          | 0.7770***<br>(0.0477) |                                                                                               |                        |                        |                        |                        |
| $\Delta_4 LTG_{it}$ |                       | 0.3134***<br>(0.0582)                                                                         | 0.2066***<br>(0.0625)  | 0.0775*<br>(0.0432)    | 0.0544***<br>(0.0183)  | 0.0038<br>(0.0251)     |
| $\widehat{FE}_{it}$ |                       | -0.1021***<br>(0.0195)                                                                        | -0.1218***<br>(0.0323) | -0.1963***<br>(0.0384) | -0.2081***<br>(0.0395) | -0.1514***<br>(0.0375) |
| AR <sup>2</sup>     | 0.02                  | -0.03                                                                                         | -0.03                  | -0.03                  | -0.03                  | -0.03                  |
| N                   | 146,151               | 133,545                                                                                       | 132,166                | 131,122                | 130,213                | 129,461                |
| Firm FE             | Y                     | Y                                                                                             | Y                      | Y                      | Y                      | Y                      |
| Time FE             | Y                     | Y                                                                                             | Y                      | Y                      | Y                      | Y                      |

# Roadmap

1. Motivation: Shiller's Volatility Puzzle
2. Expectations-Based Price Index
3. LTG and the Financial Cycle
4. LTG and Boom-Bust Investment Cycles
5. LTG and the Business Cycle

# Impulse Projections of Business Variables

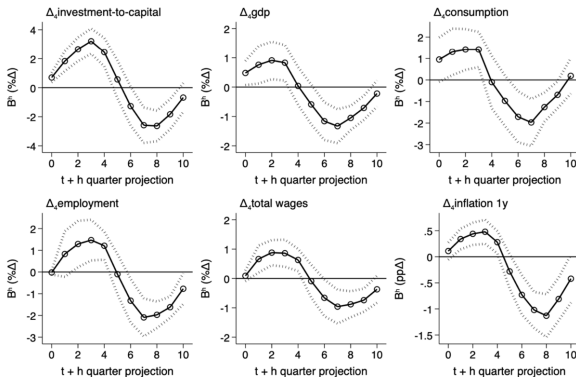
$$y_t = B_h \Delta_4 \text{LTG}_t + \mathbf{X}_t + \epsilon_t$$

- Local Projections: IRF to a one SD upward LTG revision
- LTG predicts booms and busts in other major business cycle variables
- LTG Granger causes these variables; the reverse is almost never the case

Data Description

Data Summary

Granger Test



# LTG vs Other Macro Shocks

Large body of work in macro: traces aggregate co-movement to the transmission of shocks

- Estimated using DSGE models or VARs with identifying restrictions
- Shortcoming: business cycle variation attributed to "black-box" drivers with statistical information but not interpretable

## LTG

- May be outperformed statistically, but
- Offers a source of co-movement that is directly measured at the micro-level of firms and is clearly interpretable

# Marginal Efficiency of Investment (MEI) Shocks

- Keynes coined the term to describe firms' propensity to invest, which is driven by ease of credit and "animal spirits"
- Justinano et al. (2011) formalize MEI as the productivity with which investment goods are transformed into capital
  - ▶ They estimated MEI shock using a DSGE model and found it accounts for 60-85% of US post-war fluctuations in GDP growth, hours, and investment
  - ▶ They also show MEI is high during times of easy credit as measured by low credit spreads

# Predicting MEI Shocks with LTG and Credit Spreads

This paper: predict current and future MEI shocks using

1. Current LTG revision ("good news" effect)
2. Current LTG over-optimism (predictable future disappointment)
3. Credit spreads (to account for MEI's impact on financial markets independent of LTG)

$$mei_{t+h} = B_h \Delta_4 LTG_t + \Phi_h BaaSpread_{t+h} + \delta_h \hat{FE}_h + \epsilon_{t+h}$$

- Conducted similar exercise conducted for "risk shocks" (Christiano, Motto, Rostagno 2014) and obtained similar results

# Predicting MEI shocks with LTG and Credit Spreads

- Upward LTG revisions appear as good shocks in the short term; high LTG optimism is associated with bad MEI shocks in the future
- Conditional on long term expectations, credit spreads loses its contemporaneous explanatory power for MEI
- Author's view: estimated MEI shocks do not reflect genuine bad news, but rather capture systematic disappointment of excess optimism

|                                                                                                                                                | Time Horizon of Dependent Variable (Quarters) |                     |                     |                     |                     |                    |                     |                    |                   |                   |                  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|---------------------|---------------------|---------------------|---------------------|--------------------|---------------------|--------------------|-------------------|-------------------|------------------|--|
|                                                                                                                                                | 0                                             | 1                   | 2                   | 3                   | 4                   | 5                  | 6                   | 7                  | 8                 | 9                 | 10               |  |
| Estimates From:<br>$mei_{t+h} = B_h \Delta_4 LTG_t + \emptyset_h BaaSpread_{t+h} + \delta_h \widehat{FE}_t + \varepsilon_{t+h}$<br>No Controls |                                               |                     |                     |                     |                     |                    |                     |                    |                   |                   |                  |  |
| $\Delta_4 LTG_t$                                                                                                                               | 0.19***<br>[ 0.07]                            | 0.22***<br>[ 0.07]  | 0.13<br>[ 0.08]     | 0.07<br>[ 0.07]     | 0.06<br>[ 0.06]     | 0.02<br>[ 0.07]    | 0.06<br>[ 0.06]     | -0.01<br>[ 0.07]   | -0.03<br>[ 0.07]  | -0.05<br>[ 0.07]  | -0.08<br>[ 0.09] |  |
| $BaaSpread_{t+h}$                                                                                                                              | 0.03<br>[ 0.11]                               | 0.19*<br>[ 0.11]    | 0.14<br>[ 0.09]     | 0.06<br>[ 0.08]     | 0.10<br>[ 0.09]     | -0.01<br>[ 0.09]   | 0.08<br>[ 0.08]     | 0.00<br>[ 0.08]    | -0.00<br>[ 0.07]  | 0.01<br>[ 0.07]   | -0.03<br>[ 0.07] |  |
| $\widehat{FE}_t$                                                                                                                               | -0.11**<br>[ 0.05]                            | -0.15***<br>[ 0.05] | -0.15***<br>[ 0.05] | -0.13***<br>[ 0.05] | -0.14***<br>[ 0.05] | -0.11**<br>[ 0.05] | -0.12***<br>[ 0.05] | -0.10**<br>[ 0.05] | -0.08*<br>[ 0.05] | -0.08*<br>[ 0.04] | -0.05<br>[ 0.05] |  |
| AR <sup>2</sup>                                                                                                                                | 0.02                                          | 0.06                | 0.04                | 0.03                | 0.04                | 0.02               | 0.03                | 0.01               | 0.01              | 0.01              | 0.00             |  |
| N                                                                                                                                              | 95                                            | 95                  | 95                  | 95                  | 95                  | 95                 | 95                  | 95                 | 95                | 95                | 95               |  |



# Conclusion

- Evidence on Keynes' intuition: the connection between financial markets and the macroeconomy is due to the influence of volatile, non-rational long-term expectations on both
- LTG as a simple, theory-based, and measurable "miracle" variable that:
  1. helps account for the volatility of equities and of safe / risky bonds
  2. helps explain boom-bust cycles in investment and economic activity
  3. does so through predictable disappointment of optimism
- Cannot prove that excess volatility in beliefs is *the* cause of the cycles, but the data indicate that this possibility must be seriously considered, if not adopted as a working hypothesis
- Future lines of work: theoretical macro models with overreacting beliefs; expectation formations, especially about the long term; incorporating memory and attention based theories of beliefs

Future Directions





## Comment by Venky Venkateswaran

- The extent to which the "component of LTG that is distinct from the information in prices" drives volatility and investment effects is less clear
- It would be nice to see evidence that agency frictions bias is time invariant
- It might be helpful to use the fitted forecast error specification for other results in the paper (e.g. interest rate and credit spread effects).
- Interpreting the results as the discovery of a new factor driving financial and macro volatility ("miracle variable")?
  - ▶ At the very least, it remains to be shown that LTG fluctuations can quantitatively account for the business cycle
  - ▶ It is not clear how we can distinguish LTG from other contenders (e.g. MEI shocks, risk shocks, confidence shock), especially if LTG partly reflects these factors

# Discussion by George-Marios Angeletos

Overall: impressive paper, full of interesting, informative, and novel facts

- Some challenges on structural interpretation:
  - ▶ Can we estimate the parameters  $\rho$  and  $\theta$ , which regulate the gap between subjective and rational beliefs, on some key moments and then evaluate the model's quantitative performance on other relevant moments?
  - ▶ Are the fluctuations in LTG and asset prices the product of significant over-extrapolation from past earnings? Or are they merely the manifestation of large, persistent, waves of optimism and pessimism with little connection to actual earnings?
- VAR exercise: LTG could be a proxy for at least three different forces
  1. Over-extrapolation of innovations in actual earning (author's main point)
  2. News about future TFP
  3. Extrinsic sentiments
- Hope for structural interpretation of the relevant empirical findings and a quantitative evaluation of the macroeconomic implications

# Related Literature I: Expectations and Asset Prices

Recent behavioral work combining expectations and asset price data:

1. Stock returns and bond risk premia are extrapolative
2. A line of research on expectations of future fundamentals, in particular LTG
  - ▶ **LaPorta (1996)**: introduced LTG and showed its variation predicts stock returns
  - ▶ **Bordalo, Gennaioli, LaPorta, and Shleifer 2019 (BGLS 2019)**: account for this fact using a model of diagnostic expectations
  - ▶ **BGLS 2022**: in aggregate stock market, LTG jointly predicts forecast errors and returns, and systematic changes in LTG account for the predictive power of the price-dividend ratio for returns
  - ▶ **This paper**: show expectations data also resolve Shiller's excess volatility puzzle and link LTG fluctuations to interest rates, credit spreads, and the broader business cycle

## Related Literature II: Fluctuations in Investment and Economic Activity

- Barro 1990, Fazzari et al 1988, Morck et al 1990, Lamont 1990: link the stock market to investment based on Tobin's Q
  - ▶ Stock returns predict firm level investment better than estimates of Q
- **Gennaioli, Ma, and Shleifer (2016)**: CFO optimism about 1-year ahead profits spurs firm-level investment, dwarfing the role of stock returns
- **This paper**: Focus on long-term expectations, connect investment to excess stock market volatility
  - ▶ Other papers on the role of expectations and news in the business cycle estimates shocks from VAR that is built to maximize explanatory but not easily interpretable
  - ▶ This paper: underscores the importance and promise of using a transparent measure of expectations that links between overreacting beliefs and aggregate volatility

## Related Literature III: Relaxing Rationality in Macro

- **Rational inattention / frictions** (Gabaix 2019, Angeletos et al 2020, Angeletos and Lian 2016, 2022, 2023)
- **Overreaction** (BGST 2023, Bianchi et al 2023, L'Huillier et al 2023, Maxted 2023)
- **Learning from Extreme Events** (Kozlowski et al 2019, 2020)
- **Bordalo, Gennaioli, Schleifer, and Terry (BGST 2023)**: structually estimate an RBC model with diagnostic expectations using data on CFO earnings forecasts
  - ▶ Overreaction of CFO expectations plays a quantitatively important role in driving investment at the firm level by shaping both the demand and supply of funds
- **This paper**: explicitly connect financial markets (excessively volatile) to the clear benchmark of recurrent economic fluctuations



# Future Work: Integrating survey data and realistic models of expectation formation into macro

- Microfound how non-rational expectations affect the propagation mechanism
  - ▶ Recent attempts: Angeletos et al. (2020), BGST (2023), Bianchi et al (2023), Ilut and Schneider (2014), L'Huillier et al (2023), Maxted (2023)
- Measure and study long-term expectation formation
  - ▶ Bordalo, Gennaioi, Kwon, and Schleifer (2021): show how diagnostic expectations about others may account for asset bubbles
  - ▶ Bastianello and Frontanier (2022): consider wrong beliefs about the information used by others
- What drives overreaction and why is it more prevalent in long-term expectation?
  - ▶ Keynes: long term is so uncertain and hard to imagine, these expectations are likely shaped by easily accessible current events - consistent with research in psychology.
- Psychology of memory and attention

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# Construction of the Aggregate LTG (From BGLS 2022)

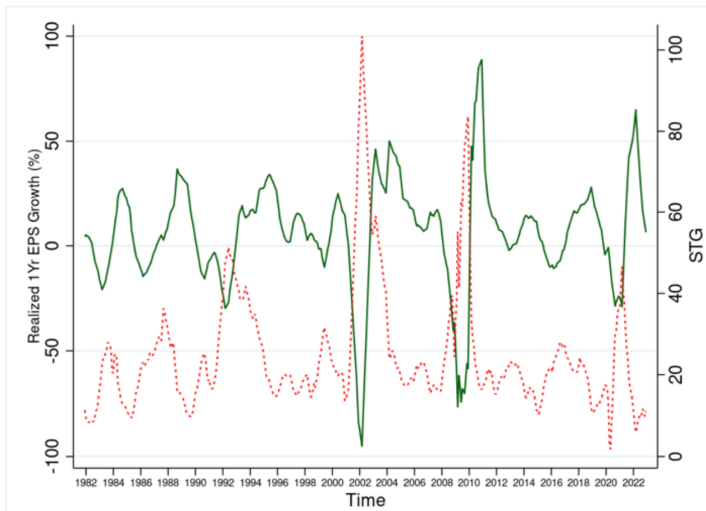
- Aggregate  $LTG_{it}$  forecasts by value-weighting firm level forecasts:

$$LTG_t = \sum_{i=1}^S LTG_{i,t} \frac{P_{i,t} Q_{i,t}}{\sum_{i=1}^S P_{i,t} Q_{i,t}}$$

where  $S$  is the number of firms in the  $S\&P$  index with IBES data on  $LTG_{i,t}$ ,  $P_{i,t}$  and  $Q_{i,t}$  are the stock price and number of shares outstanding of firm  $i$  at time  $t$ , respectively.

- Note: It is not clear whether  $LTG$  captures the geometric or arithmetic average; the authors take the former interpretation but argues the distinction is not key for studying return predictability.

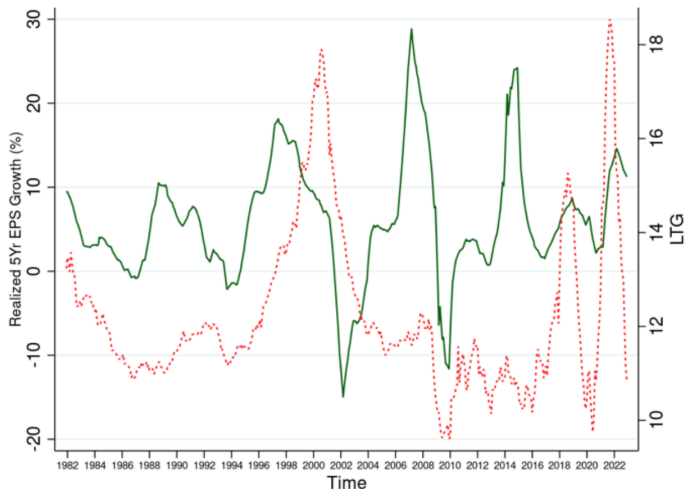
# 1-year Ahead Earnings Growth and Expectations



Red: STG; Green: realized 1-year earnings growth

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# 5-year Ahead Earnings Growth and Expectations



Red: LTG; Green: realized 5-year earnings growth

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# Summary Statistics for Measures of Expectations

|                                  | Date Range | N   | Mean  | St.D  | Min    | Max    |
|----------------------------------|------------|-----|-------|-------|--------|--------|
| STG1                             | 1976-2021  | 181 | 19.52 | 11.45 | 0.37   | 88.03  |
| $\Delta_4$ STG1                  | 1977-2021  | 175 | 0.62  | 14.96 | -67.25 | 66.54  |
| $STG1_t - \Delta_4 e_{t+4}$      | 1976-2020  | 175 | 15.31 | 27.06 | -27.52 | 140.84 |
| LTG                              | 1981-2021  | 161 | 13.28 | 2.20  | 10.25  | 21.87  |
| $\Delta_4$ LTG                   | 1981-2021  | 157 | 0.14  | 1.92  | -5.10  | 9.31   |
| $LTG_t - \Delta_{20} e_{t+20}/5$ | 1982-2017  | 141 | 5.66  | 8.46  | -21.21 | 26.95  |
| Expected Returns CFO             | 1991-2022  | 128 | 5.27  | 1.21  | 2.20   | 9.10   |
| Expected Returns SPF             | 1991-2022  | 128 | 7.68  | 5.03  | -2.95  | 22.03  |
| Expected Returns AAI             | 1987-2022  | 142 | 6.70  | 13.09 | -29.15 | 39.03  |
| Expected Returns PCA             | 1991-2022  | 128 | 0.02  | 1.54  | -3.30  | 4.98   |

## Descriptions for Measures of Expectations

- **LTG**: aggregate market expectations for 5-year EPS growth, calculated by value weighting firm level forecasts
- **$\Delta_4\text{LTG}_t$** : 4-quarter percentage point change in aggregate market expectation for 5-year EPS growth, calculated by value weighting firm level forecasts
- **$\text{PLTG}_t - \Delta_{20}\text{e}_{t+20}/5$** : the percentage point difference in 5 year forecasted growth in earnings and realized 5-year earnings growth
- **STG<sub>1</sub>**: aggregate market expectations for 1-year EPS growth, calculated by value weighting firm level forecasts
- **$\Delta_4\text{STG}_t$** : 4-quarter percentage point change in aggregate market expectation for 1-year EPS growth
- **$\text{STG}_t - \Delta_4\text{e}_{t+4}$** : the percentage point difference in 1 year forecasted growth in earnings and realized 1-year earnings growth

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## Descriptions for Measures of Expectations

- **Expected Returns CFO**: average expectation of 1-year returns on the S&P 500 from Richmond Fed's CFO survey (spans Q4 2001-2022, with missing values from Q1 1991-Q32002 backfilled by taking fitted value from a regression of ER CFO on the first principal component of ER SPF and ER AAI)
- **Expected Returns SPF**: average expectation of 1-year returns on the S&P 500 from the Survey of Professional Forecasters, where missing values are carried forward.
- **Expected Returns AAI**: % of bullish - % bearish investors from the American Association of Individual Investors.
- **Expected Returns PCA**: the first principal component for the three measures of expected returns

The results from Rationality Test 2 using Expected Returns CFO also holds using these alternative measures of expected returns, which is shown in Table A5 in the Appendix.

# Summary Statistics for Financial and Macro Variables

|                                | Date Range | N   | Mean  | St.D | Min   | Max   |
|--------------------------------|------------|-----|-------|------|-------|-------|
| $\Delta$ tbill 1Y              | 1963-2022  | 239 | -0.01 | 1.71 | -6.54 | 6.60  |
| $\Delta$ tbill 10Y             | 1963-2022  | 239 | -0.02 | 1.23 | -4.29 | 3.97  |
| $\Delta$ baa credit spread 10Y | 1963-2022  | 239 | 0.02  | 0.78 | -3.49 | 3.39  |
| $\Delta$ gdp                   | 1948-2022  | 300 | 0.03  | 0.03 | -0.09 | 0.13  |
| $\Delta$ consumption           | 1960-2022  | 252 | 0.03  | 0.03 | -0.17 | 0.22  |
| $\Delta$ investment-to-capital | 1948-2022  | 300 | -0.00 | 0.07 | -0.26 | 0.23  |
| $\Delta$ unemployment rate     | 1949-2022  | 296 | 0.00  | 1.55 | -8.60 | 11.10 |
| $\Delta$ employment            | 1949-2022  | 296 | 0.01  | 0.02 | -0.16 | 0.13  |
| $\Delta$ total wages           | 1960-2022  | 252 | 0.03  | 0.03 | -0.07 | 0.10  |
| inflation 1y                   | 1948-2022  | 300 | 3.53  | 2.95 | -2.87 | 14.59 |
| $\Delta$ sp500                 | 1976-2021  | 184 | 0.08  | 0.15 | -0.51 | 0.42  |
| $\Delta$ federal funds rate    | 1955-2022  | 270 | 0.00  | 2.08 | -6.82 | 10.01 |



# Descriptions for Interest Rates and Credit Spreads

- $\Delta_4$  tbill 1y is the 4-quarter percentage change in the Federal Reserve's 1-year treasury bond yield (DGS1)
- $\Delta_4$  tbill 10y is the 4-quarter percentage change in the Federal Reserve's 10-year treasury bond yield (DGS10)
- $\Delta_4$  baa credit spread 10Y is the 4-quarter percentage change in the yield spread between Moody's 10y BAA bond (BAA) and the US 10-year Treasury Bond (DGS10).

From the abbreviations, the data seems to be all from FRED.

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## Descriptions for Macro Variables

- **$\Delta_4$  gdp** is the 4-quarter log change in real GDP (GDP / GDPDEF)
- **$\Delta_4$  consumption** is the 4-quarter log change in real consumption (PCE / GDPDEF)
- **$\Delta_4$  investment-to-capital** is the 4-quarter log change in the ratio of non-residential investment (PNFI) to the previous year's cost of capital stock (K1NTOTL1ES000)
- **$\Delta_4$  unemployment rate** is the 4-quarter percentage point change in the unemployment rate (UNRATE)
- **$\Delta_4$  employment rate** is the 4-quarter log change in total employment (CE160V)
- **$\Delta_4$  total wages** is the 4-quarter log change in total real wage disbursements (A576RC1 / GDPDEF)
- **inflation 1y** is the 4-quarter percentage point change in annual CPI inflation (CPIAUCSL)
- **$\Delta_4$  federal funds rate** is the 4-quarter percentage point change in the policy interest rate (FEDFUNDS).

From the abbreviations, the data seems to be all from FRED.

[Back to Macro](#)

# Key Role of Long-term Expectations

- De La O and Myers (2021): short-term earnings growth expectations help account for variation in the price-dividend ratio
- BGLS (2022): While short term expectations are fairly accurate, LTG exhibits a marked departure from rationality in the form of overreaction or excess volatility
  - ▶ High LTG and increases in LTG predict disappointment of earnings growth expectations at a 3-5 year horizon
  - ▶ High current LTG predicts future low stock returns, while short term earnings expectations do not
- This paper: LTG captures slow moving forecasts of long-term growth. Long term beliefs are difficult to form and may help explain market movements

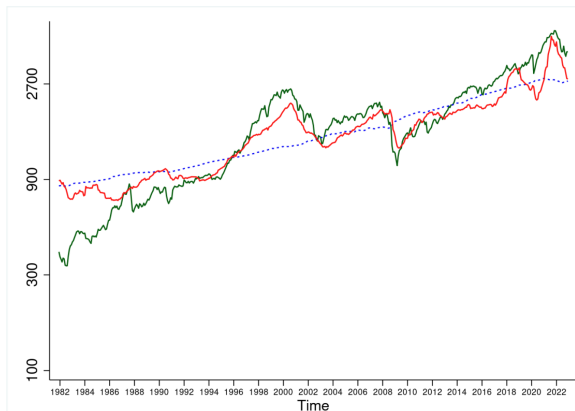
# Predictability of Short-term Forecast Errors with Short Term Expectations

|                   | Time Horizon of Dependent Variable (Quarters)                                               |         |         |         |         |         |         |         |         |         |       |
|-------------------|---------------------------------------------------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|-------|
|                   | Estimates From: $STG1_{t+h} - \Delta_4 e_{t+h+4} = B_h \Delta_4 STG1_t + \varepsilon_{t+h}$ |         |         |         |         |         |         |         |         |         |       |
|                   | 0                                                                                           | 1       | 2       | 3       | 4       | 5       | 6       | 7       | 8       | 9       | 10    |
| $\Delta_4 STG1_t$ | 2.99                                                                                        | 1.39    | -0.28   | -1.21   | -1.21   | -0.93   | -0.76   | -0.93   | -1.40   | -1.56   | -1.85 |
| [ 2.90]           | [ 2.18]                                                                                     | [ 1.68] | [ 1.40] | [ 1.25] | [ 1.12] | [ 0.98] | [ 0.91] | [ 0.91] | [ 1.08] | [ 1.24] |       |

The estimates measure the impact of a 1 standard deviation change in revisions to short-term earnings growth expectations,  $\Delta_4 STG1_t$ , on forecast errors for expected short term growth.  $\Delta_4 STG1_t$  is the 4-quarter percentage point change in aggregate market expectation for 1-year earnings per share growth, calculated by value weighting firm level forecasts.  $STG1_{t+h} - \Delta_4 e_{t+h+4}$  is the percentage point difference in one year forecasted growth in earnings and realized 1-year earnings growth. All regressions are unconditional. Heteroskedasticity-consistent asymptotic standard errors reported in parentheses are computed according to Huber-White. \*, \*\*, and \*\*\* indicate statistical significance at the 90%, 95%, and 99%, confidence level, respectively.

# Expectations-Based Price Index, Adjusted for Inflation

- Adjusted for inflation: *SP500* index (green), rational benchmark (blue), eexpectations-based price (red).



# Quantitative Drawbacks of the Minimal Model

- This model captures minimal features of belief overreaction; missing ingredients that are important to quantitatively match overreaction in the data
- 1. Here, investors only overreact to tangible cash flow news
  - ▶ BGLS (2022) studies overreaction to intangible news about future prospects
- 2. Missing production side, which is needed to quantitatively assess belief overreaction and aggregate investment
  - ▶ BGST (2021) build and estimate an RBC model using measured CFO forecasts to show the importance of belief overreaction for credit and investment cycles

# LTG and Cumulative Changes in Investment-to-Capital

**Table B.8:** LTG and Cumulative Changes in investment-to-capital

|                                                                                                                                        | Time Horizon of Dependent Variable (Quarters) |         |         |         |          |          |          |          |          |          |          |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|---------|---------|---------|----------|----------|----------|----------|----------|----------|----------|
|                                                                                                                                        | 0                                             | 2       | 4       | 6       | 8        | 10       | 12       | 14       | 16       | 18       | 20       |
| Estimates From: $\Delta_h \text{investment-to-capital}_{t+h} = B_h \Delta_4 LTG_t + \delta_h \widehat{FE}_t + X_t + \varepsilon_{t+h}$ |                                               |         |         |         |          |          |          |          |          |          |          |
| First Stage: $LTG_t - \Delta_{20} e_{t+20}/5 = \Phi LTG_t + \varepsilon_t \rightarrow \widehat{FE}_t$                                  |                                               |         |         |         |          |          |          |          |          |          |          |
| $\widehat{FE}_t$                                                                                                                       | 0.13                                          | 0.25    | -0.31   | 1.45*** | -2.29*** | -3.06*** | -3.78*** | -4.37*** | -4.87*** | -5.26*** | -5.55*** |
|                                                                                                                                        | [ 0.30]                                       | [ 0.42] | [ 0.58] | [ 0.67] | [ 0.71]  | [ 0.67]  | [ 0.68]  | [ 0.68]  | [ 0.76]  | [ 0.84]  | [ 0.89]  |
| AR <sup>2</sup>                                                                                                                        | 0.03                                          | 0.46    | 0.39    | 0.28    | 0.26     | 0.30     | 0.32     | 0.32     | 0.33     | 0.28     | 0.21     |
| N                                                                                                                                      | 138                                           | 138     | 138     | 138     | 138      | 138      | 138      | 138      | 138      | 138      | 138      |

# LTG and Investment with Firm and Time FE

- Firm-level regressions for all US firms in the IBES sample from 1982 Q4 to 2018 Q1.
- Firm-level forecast error  $FE_{it}$ : difference between  $LTG_{i,t}$  and average annual growth rate in firm  $i$ 's EPS  $\Delta_{20} e_{i,t+20}/5$ .
- Firm  $i$ 's investment  $i_{i,t}$  is defined to be the log of  $\Delta_4 K_{i,t+h}/K_{i+h-4}$ , where firm  $i$ 's capital stock  $K_{i,t}$  includes physical, intangible and knowledge capital following the methodology of Peters and Taylor (2017).
- First Stage: OLS regression of  $FE_{i,t}$  on  $LTG_{i,t}$ .
- Second stage:  $\Delta_4 i_{i,t+h} = B_h \Delta_4 LTG_{it} + \delta_h \hat{FE}_{it} + \epsilon_{t+h}$ .

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# Granger Causality Test: LTG and Expected Returns

- Estimate VAR systems that include one of four measures expected returns, LTG, along with the policy rate, CPI inflation, S&P 500 yearly returns
- LTG strongly predict expected returns ( $L \implies ER$ ), but the inverse ( $ER \implies L$ ) is only true in higher lag specifications

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VAR system: [Expected Returns, LTG, Controls]

$$ER_t = \sum_1^h A_h ER_{t-h} + \sum_1^h B_h LTG_{t-h} + \sum_1^h C_h X_{t-h} + \varepsilon_t$$

$$LTG_t = \sum_1^h A_h ER_{t-h} + \sum_1^h B_h LTG_{t-h} + \sum_1^h C_h X_{t-h} + \varepsilon_t$$

|                           | Lags (h) = 2       |                    | Lags (h) = 4       |                    | Lags (h) = 8       |                    |
|---------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| Returns Expectation       | ER $\Rightarrow$ L | L $\Rightarrow$ ER | ER $\Rightarrow$ L | L $\Rightarrow$ ER | ER $\Rightarrow$ L | L $\Rightarrow$ ER |
| (1) Expected Returns CFO  | 0.73               | <b>0.06</b>        | 0.19               | <b>0.01</b>        | <b>0.04</b>        | <b>0.00</b>        |
| (2) Expected Returns SPF  | 0.83               | <b>0.06</b>        | 0.25               | <b>0.00</b>        | <b>0.02</b>        | <b>0.00</b>        |
| (3) Expected Returns AAII | 0.71               | <b>0.01</b>        | 0.57               | <b>0.01</b>        | 0.20               | <b>0.04</b>        |
| (4) Expected Returns PCA  | 0.91               | 0.15               | 0.20               | <b>0.01</b>        | <b>0.08</b>        | <b>0.00</b>        |

Four measures of expected one-year SP500 returns, using CFO surveys, Survey of Professional Forecasters (SPF), American Association of Individual Investors (AAII), and the first principal component for the three measures (PCA).

# Granger Causality Test: LTG and Macro Variables

- Estimate VAR systems that include LTG and one of the six real variables.
- Null hypothesis for  $L \implies R$ : LTG does not predict the real variable
- Null hypothesis for  $R \implies L$ : the real variable does not predict LTG

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VAR system: [Real Variable, LTG]

$$y_t = \sum_1^h A_h y_{t-h} + \sum_1^h B_h LTG_{t-h} + \varepsilon_t$$

$$LTG_t = \sum_1^h A_h y_{t-h} + \sum_1^h B_h LTG_{t-h} + \varepsilon_t$$

| Real Variable             | Lags (h) = 2      |                   | Lags (h) = 4      |                   | Lags (h) = 8      |                   |
|---------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                           | R $\Rightarrow$ L | L $\Rightarrow$ R | R $\Rightarrow$ L | L $\Rightarrow$ R | R $\Rightarrow$ L | L $\Rightarrow$ R |
| (1) gdp                   | 0.67              | <b>0.00</b>       | 0.93              | <b>0.00</b>       | <b>0.07</b>       | <b>0.00</b>       |
| (2) investment-to-capital | 0.74              | <b>0.00</b>       | 0.46              | <b>0.00</b>       | 0.28              | <b>0.04</b>       |
| (3) consumption           | 0.43              | <b>0.01</b>       | 0.82              | <b>0.00</b>       | <b>0.01</b>       | <b>0.00</b>       |
| (4) employment            | 0.11              | <b>0.01</b>       | 0.16              | <b>0.00</b>       | <b>0.00</b>       | <b>0.00</b>       |
| (4) unemployment rate     | 0.18              | <b>0.00</b>       | 0.33              | <b>0.00</b>       | <b>0.00</b>       | <b>0.00</b>       |
| (5) total wages           | 0.13              | <b>0.00</b>       | 0.18              | <b>0.00</b>       | <b>0.00</b>       | <b>0.01</b>       |
| (6) inflation 1y          | <b>0.09</b>       | <b>0.01</b>       | 0.29              | <b>0.01</b>       | 0.61              | <b>0.00</b>       |